

Question 1 →

a. Steps Heuristic:

- For every mismatched letter between the current word and the goal word, a mismatch penalty of +1 is added
- The unavoidable remaining cost it lower-bounds is the steps cost associated with the number of letter replacements needed to transform the current word into the goal word
 - Every mismatched letter has to be changed at least once
 - Each of these changes is associated with a cost penalty of 1
- The Steps heuristic cannot overestimate the true optimal remaining cost because for every mismatched letter there is, each requires only one action to replace that letter

b. Scrabble Heuristic:

- For every mismatched letter between the current word and the goal word, a penalty is added based on the Scrabble values in the goal word for which the current mismatched letters have to be changed to
- The unavoidable remaining cost it lower-bounds is the Scrabble cost of changing the mismatched letters into the correct goal letters
- The Scrabble heuristic cannot overestimate the true optimal remaining cost because every mismatched letter has to be changed eventually, each associated with a corresponding scrabble letter cost based on that goal letter.

c. Frequency Heuristic:

- For every mismatched letter between the current word and the goal word, the number of mismatches is multiplied by $(1 + R_{\min})$
- $1 + R_{\min}$ represents the minimum cost of a word transition
- The unavoidable remaining cost it lower-bounds is based on the number of mismatched positions along with the minimum possible frequency cost of each required transition
- The frequency heuristic cannot overestimate the true optimal remaining cost because every mismatched letter requires at the minimum one action, and that action is associated with cost $1 + R_{\min}$.
- By multiplying the number of mismatches with that associated cost, it shows the lower bound on a remaining cost

Question 2 →

Word Pair	Cost	UCS Cost	UCS Nodes	A* cost	A* Nodes	Reduction
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	Measure		Expanded		Expanded	
WHY → ASK	Steps	5	773	5	16	97.93
WHY → ASK	Scrabble	9	765	9	13	98.30
WHY → ASK	Frequency	38.86	808	38.86	754	6.68
BEAR → DUCK	Steps	4	691	4	6	99.13
BEAR → DUCK	Scrabble	11	2981	11	6	99.80
BEAR → DUCK	Frequency	38.24	1084	38.24	714	34.13

Reduction Formula:

$$100 \times \frac{\text{UCS expansions} - \text{A}^* \text{ expansions}}{\text{UCS expansions}}$$

Question 2 (A) →

- UCS and A* did always return the same optimal cost for both word pairs (WHY → ASK) and (BEAR → DUCK) across the Steps, Scrabble, and Frequency heuristic

Question 2 (B) →

- For the word pair (WHY → ASK), the largest percent reduction in expanded nodes came from Scrabble
- For the word pair (BEAR → DUCK), the largest percent reduction in expanded nodes also came from Scrabble

Question 2 (C) →

- For the word pair (WHY → ASK), the cost measure in which the the heuristic was least informative was Frequency
- For the word pair (BEAR → DUCK), the cost measure in which the heuristic was least informative was Frequency

Question 2 (D) →

- The heuristic scrabble producing the largest percent reduction is due to the fact that the heuristic is based on values of goal letters which provides more information compared to the Steps heuristic that simply looks at number of mismatches

- The frequency heuristic was the least informative for both word pairs since it is based on using the minimum rarity value across the dictionary, so its estimated costs provide less information to A^*
 - As a result, A^* has to still expand many nodes which is why the number of nodes A^* expanded was very similar to that expanded by UCS

Question 3 →

1. Steps Cost Measure

- The optimal cost from W_1 to W_2 must always equal the optimal cost from W_2 to W_1
- The number of transformations needed to reach the goal is the same regardless of moving from W_1 to W_2 or from W_2 to W_1 , and each letter mismatch obtains a penalty mismatch of +1
- For example, regardless of moving from Dog → Cat or Cat → Dog, there are still three letter mismatches present from the word to the goal word, and ultimately a penalty score of +3.

2. Scrabble Cost Measure

- The optimal cost from W_1 to W_2 will not always equal the optimal cost from W_2 to W_1
- This is because the cost is based on the letter that is being inserted, and when you do the reverse change, you end up changing the letter being added, and different letters are associated with a different cost
- For example, if you were to calculate the cost of going from Dog → Dot, the cost would be the scrabble value of T since that is the only mismatched letter, so 1 according to the dictionary of scrabble values.
- If you were to now do the inverse and go from Dot → Dog, the cost would be the scrabble value of G since that is the only mismatched letter, so 2 according to the dictionary of scrabble values

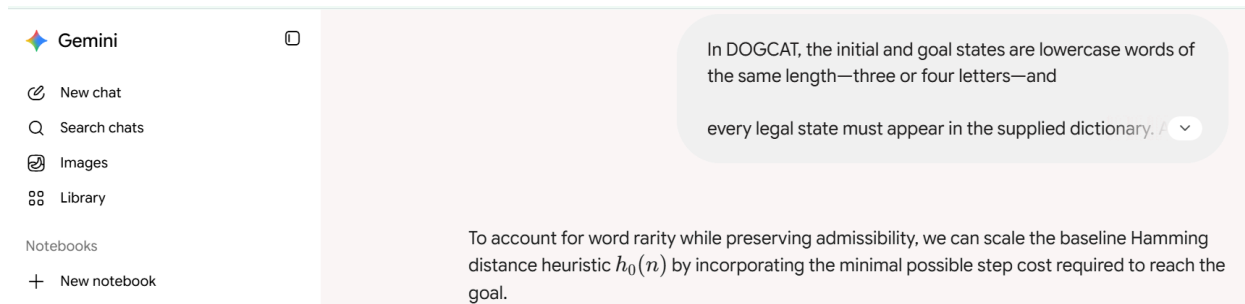
3. Frequency Cost Measure

- The optimal cost from W_1 to W_2 will not always equal the optimal cost from W_2 to W_1
- This is because the cost of a transition is based on a rarity value of the new word, and different words have different rarity values associated with them
- Using the same example of Dot and Dog, since the rarity value for Dot is different than Dog, than the cost associated with the transition from Dot → Dog would be different from the cost associated with the transition from Dog → Dot

Question 4 →

a.

- **Model Used:** Gemini 3.6 Flash
- **Exact Prompt:** Pasted into Gemini 3.6 Flash
- **Proposed Heuristic:** Scale the baseline Hamming distance heuristic ($h_0(n)$) by incorporating the minimal possible step cost required to reach the goal the minimal possible step cost required to reach the goal.



- b. Proposed Heuristic Definition: $h_1(n) = h_0(n) * (1 + R_{\min})$
- Where $h_0(n)$ is the standard baseline Hamming distance (the number of positions where n differs from g).
 - Where $R_{\min}$ be the minimum possible rarity value among all words in dictionary D
- c. The heuristic provided by Gemini 3.6 is admissible since $R_{\min}$ is greater than or equal to zero, and since every action has cost of at least $1 + R_{\min}$.
- d. After using the heuristic provided from Gemini 3.6 Flash:
- WHY → ASK: The number of nodes expanded was 754 for A^*
 - The number of nodes expanded was 808 using h_0 ,
 - The A^* cost was the same as the UCS cost
 - BEAR → DUCK: The number of nodes expanded was 716 for A^*
 - The number of nodes expanded was 1084 using h_0
 - The A^* cost was the same as the UCS cost
- e. The evidence supports using the heuristic because the amount of nodes expanded for both WHY → ASK and BEAR → DUCK was lower compared to h_0 , showcasing additional search information, and it still led to the same cost as h_0 , so there was no sacrifice in optimality.

AI was used to help understand the different heuristics.